

Contribution and prior work for Erdős 883

First question only 5 October 2026

Result to evaluate. For every n and every $A \subseteq \{1, \dots, n\}$ with $|A| > \lfloor n/2 \rfloor + \lfloor n/3 \rfloor - \lfloor n/6 \rfloor$, the coprime graph on A contains C_{2k+1} for every $1 \leq k \leq \lfloor n/6 \rfloor$. The final Lean declaration is

`Erdos883Verified.erdos883_firstQuestion.`

Erdős and Sárközy 1997. Their odd-cycle theorem establishes a positive linear range for sufficiently large n and asks whether the half-length constant can be $1/6$. This is historical problem context, not a theorem assumed in the final proof. Primary paper.

Della Pietra July 2026. The pinned manuscript states the sharp $1/6$ half-length range for all sufficiently large n . Missing-even surplus smoothing, totient profiles, prime-signature ordering and the Hall construction are central prior contributions. The present proof adapts that architecture and does not claim it as new. Pinned manuscript at publication revision 1f26276.

What this artifact adds. An exact interval criterion covers all permitted subsets, rather than sampled examples. The finite proof covers $0 \leq n \leq 199999$, including 88 adaptive intervals from 1212 onward; a formal analytic proof covers every $n \geq 200000$. Refined finite-prefix CDF certificates, exact rounding allowances and an improved universal common-neighbor estimate support that threshold. An elementary odd-triple pigeonhole argument supplies coprime endpoints without an external Ahlswede–Khachatrian assumption. The final canonical all- n theorem joins the finite and analytic ranges.

Formal evidence. Lean 4.33.1 and pinned mathlib check the final declaration and its unabbreviated canonical type. The reported assumptions are exactly `propext`, `Classical.choice`, `Quot.sound`. The frozen local closure contains 6831 modules. The recorded verification combines modular source compilation, targeted correspondence checks, and a fresh final-root/raw-type check. It is not a second clean rebuild of every numerical certificate or independent human review.

What belongs to other work. The complete-tripartite second question is separate: see Sárközy 1999 and the public second-question Lean project. Neither result is attributed to this contribution. The canonical statement source fixes the target; it is not the proof.

Attribution and readiness. This is a source-specific contribution comparison, not an exhaustive priority determination. No first-ever claim is made. AI involvement was substantial in mathematical work, formalization, certificates, checking and writing. The existing byline is retained for author review; the author’s actual personal contributions, tool/model record and independent review must be confirmed before submission. Public availability of the final artifact and external human review are not asserted.

Artifact identity. Version 5 of `Erdos883_FirstQuestion_Lean_20261005.zip`, 20359760 bytes. SHA-256:

8f017032cb93f2f0891f6f8200390c161acabf112dac72567462e642b91ede95

The companion manuscript explains the proof, and the reproduction guide separates replay instructions from the evidence already obtained.